Activity 2: Medical Microchip Failure Triage: ESD, EOS or Process Defect?

Objective: Build an evidence-based hypothesis that separates HBM/CDM exposure, EOS and non-electrical process causes.

Scenario

A medical-device ASIC passes wafer test but shows intermittent leakage after final assembly. A handler alarm occurred during the lot, yet the power-supply log also shows a transient. Management wants a defensible triage before quarantined product is released.

Roles

- reliability engineer
- failure-analysis lead
- production manager

Detailed procedure

1. Review the event chronology, product symptoms, handler alarm and power-transient records.
2. Classify each clue as product, process, equipment, environmental or personnel evidence.
3. Describe the HBM and CDM mechanisms that could fit the handling sequence.
4. Describe the EOS mechanism that could fit the power record.
5. List at least two non-ESD alternatives that must remain open.
6. Create a comparison matrix of expected evidence for catastrophic, latent/cumulative and EOS damage.
7. Select immediate containment for the lot and neighbouring lots.
8. Specify non-destructive checks before destructive failure analysis.
9. Define the samples and controls required for a credible comparison.
10. State what result would support or weaken each hypothesis.
11. Prepare a five-minute management briefing that avoids claiming ESD without proof.
12. Record the next decision gate and authorised disposition options.

Questions

- Why is a monitor alarm not proof that this device was damaged by ESD?
- Which facts make CDM plausible?
- What would make the latent-failure claim weak?

Evidence to submit

- Chronology
- Hypothesis matrix
- Containment plan
- Decision-gate briefing

Acceptance criteria

- [ ] ESD, EOS and process alternatives remain explicit
- [ ] Evidence needed for each hypothesis is stated
- [ ] Containment protects product and evidence
- [ ] Conclusions use calibrated language

Debrief

Explain which evidence changed your first assumption, which control is most reliable, and what follow-up measurement is needed.